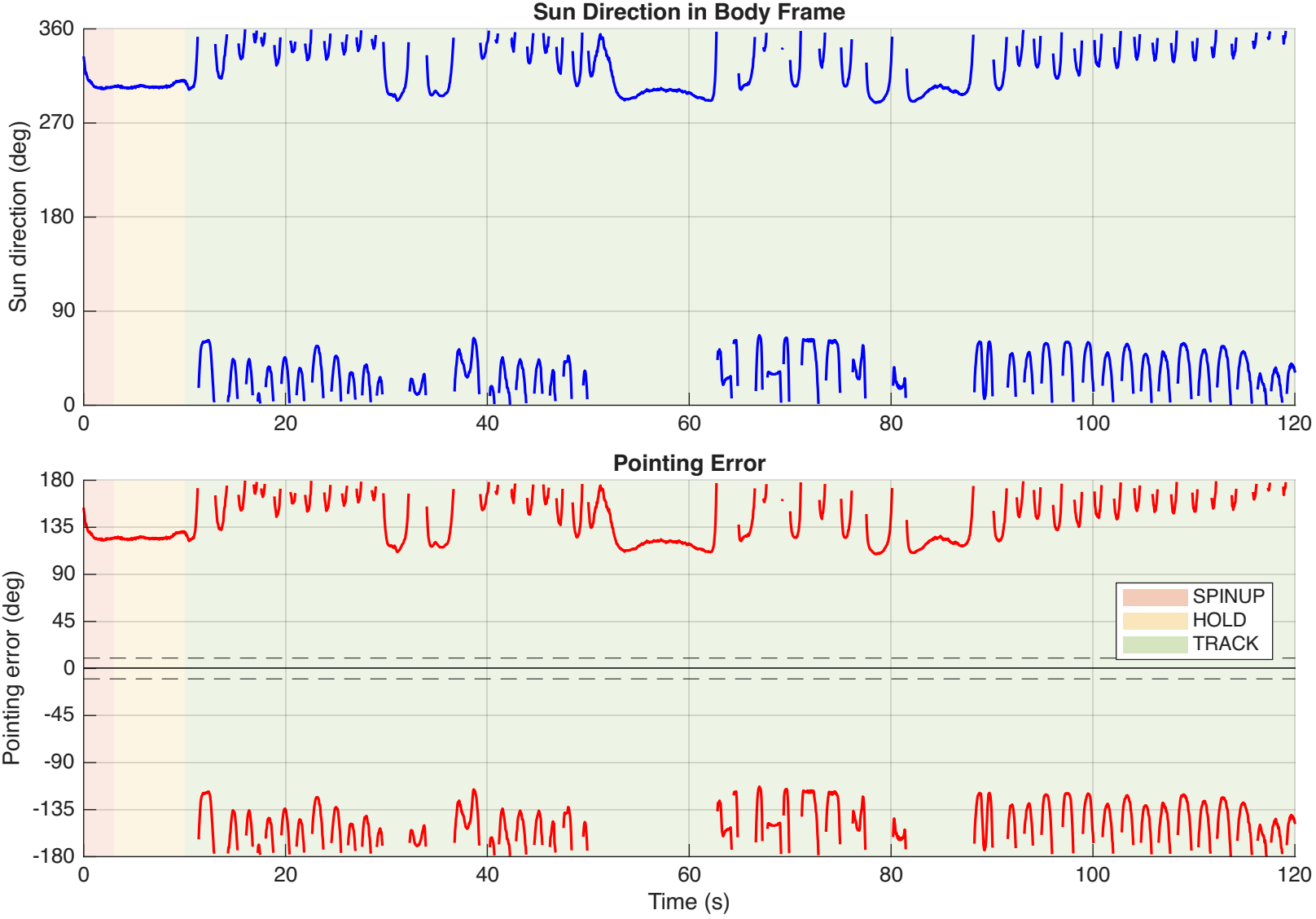
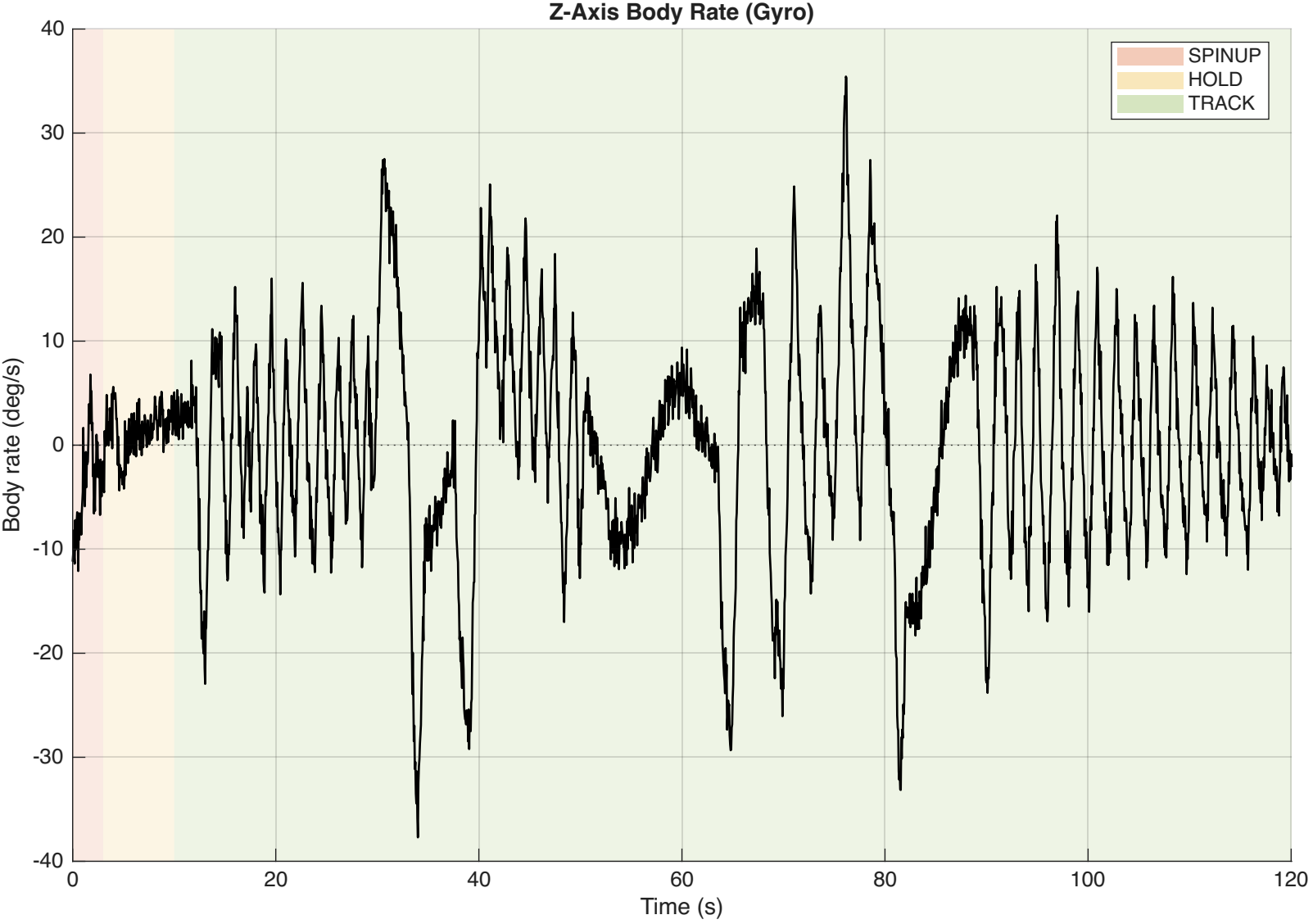
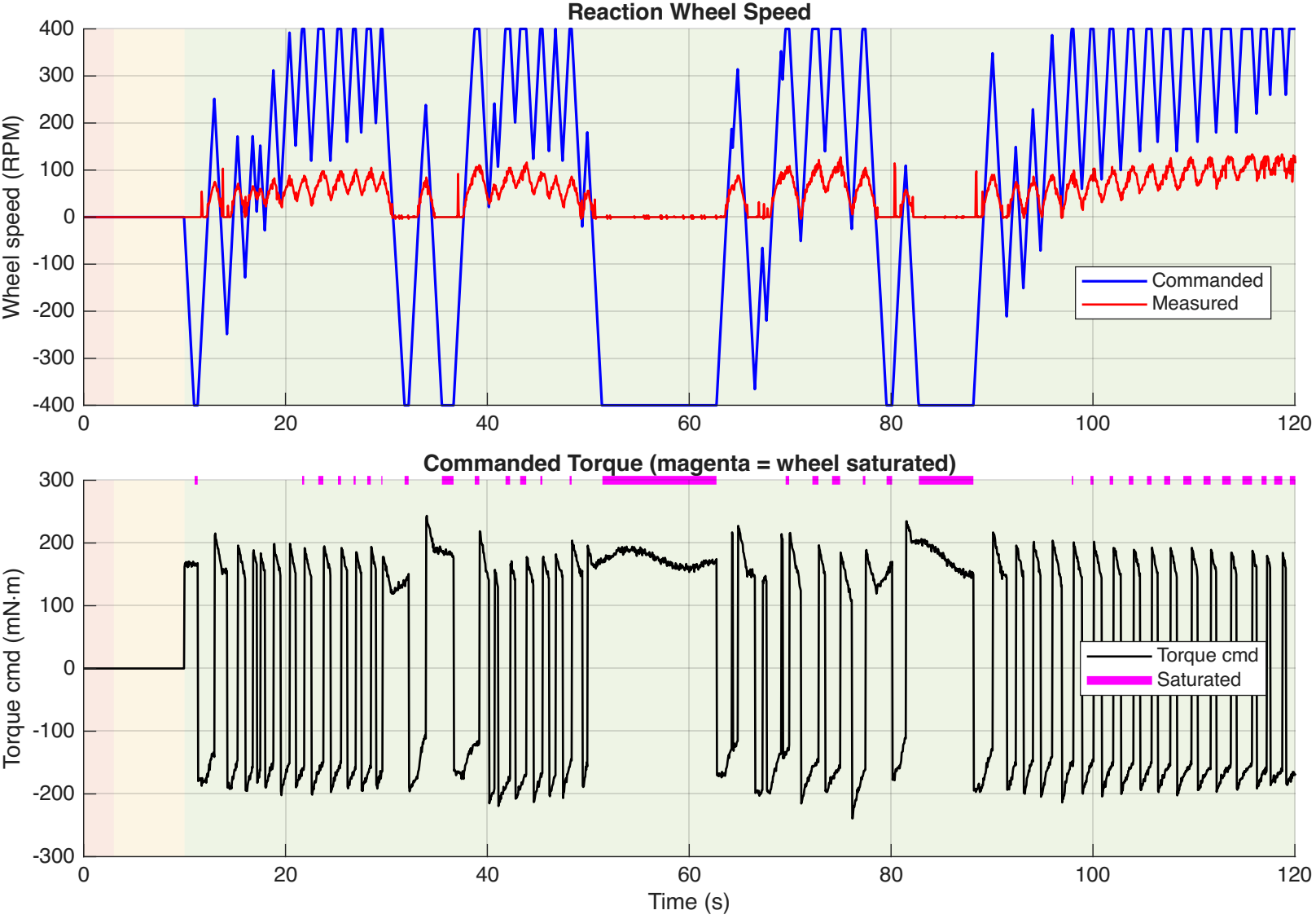
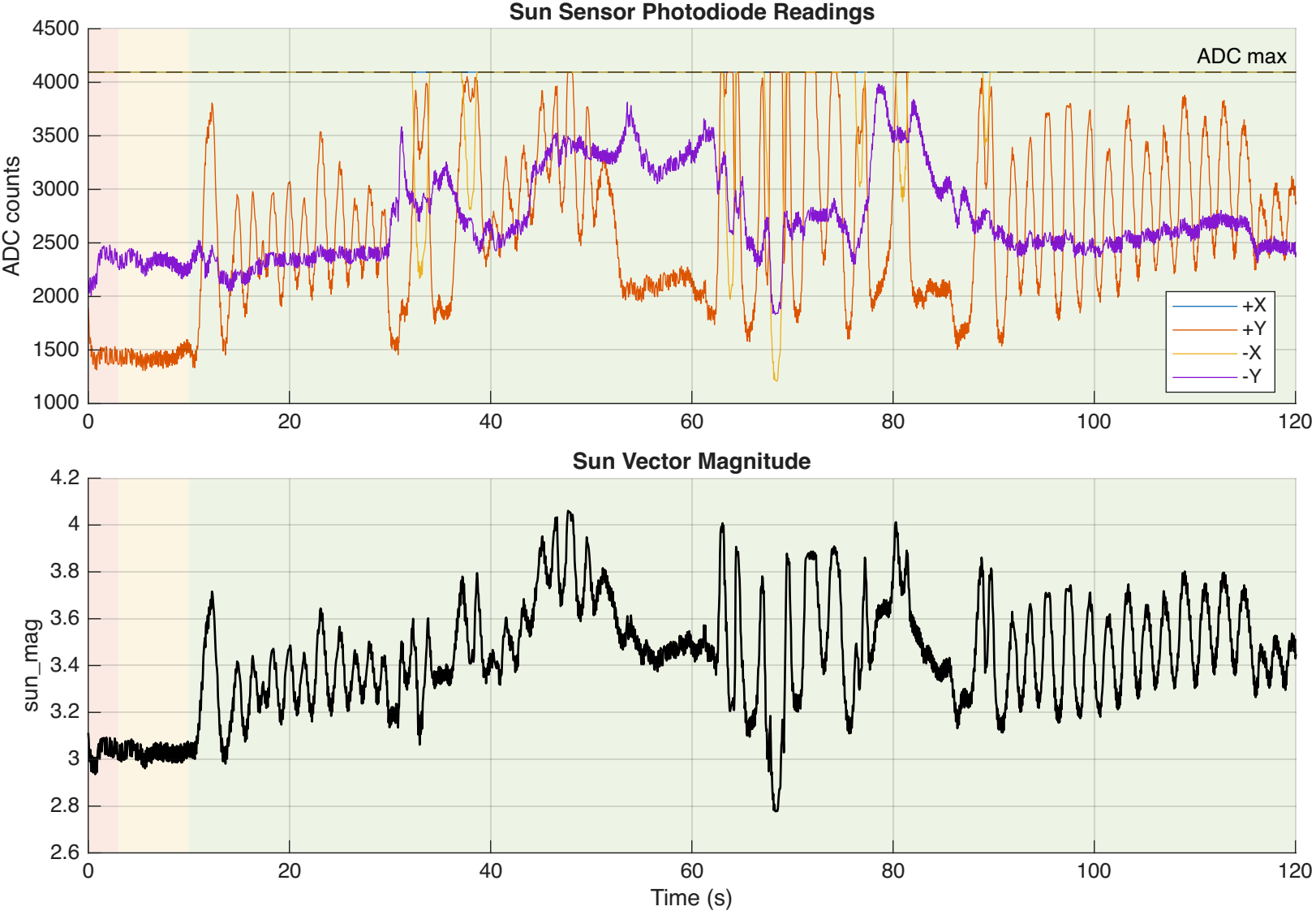
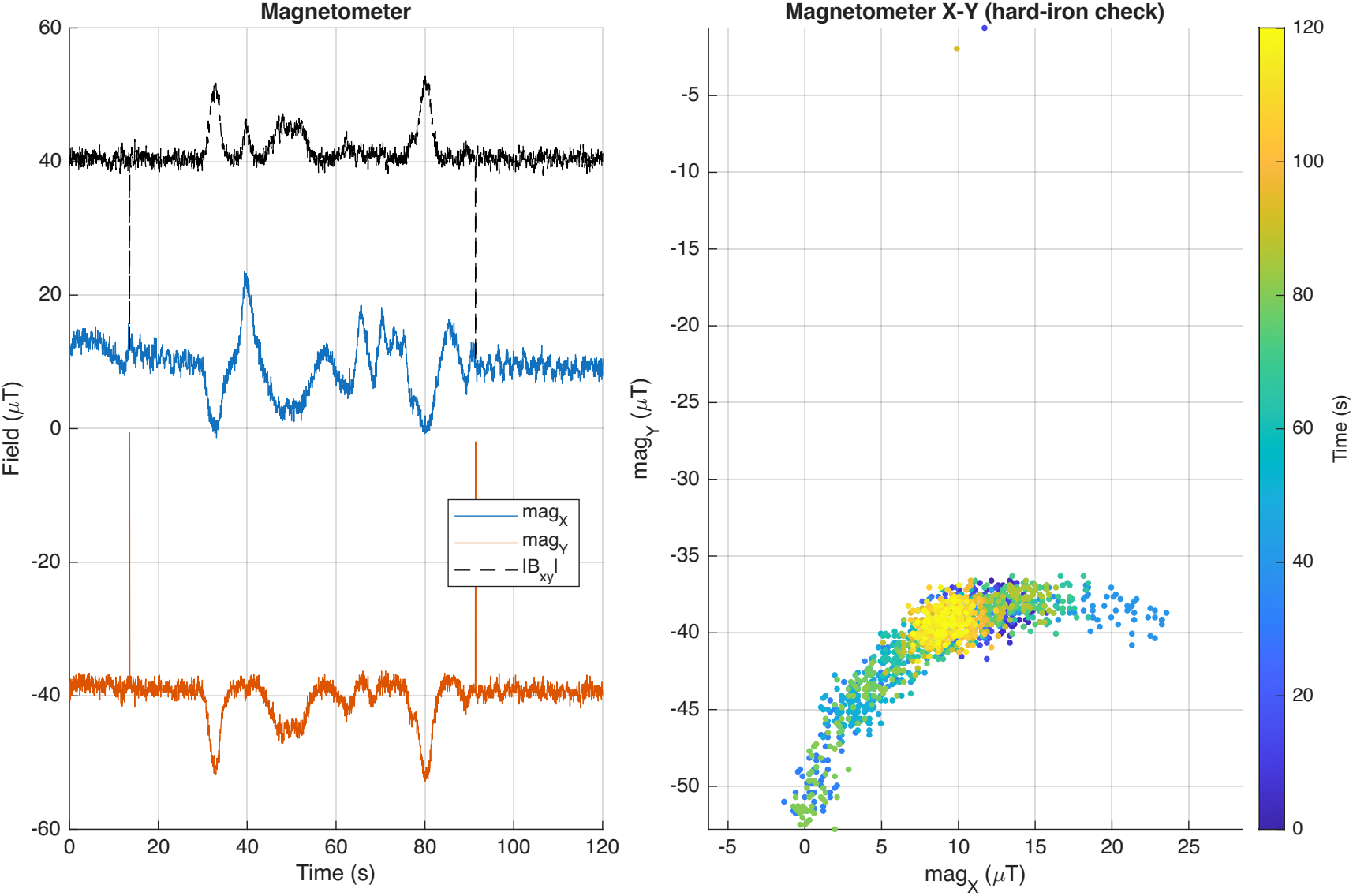
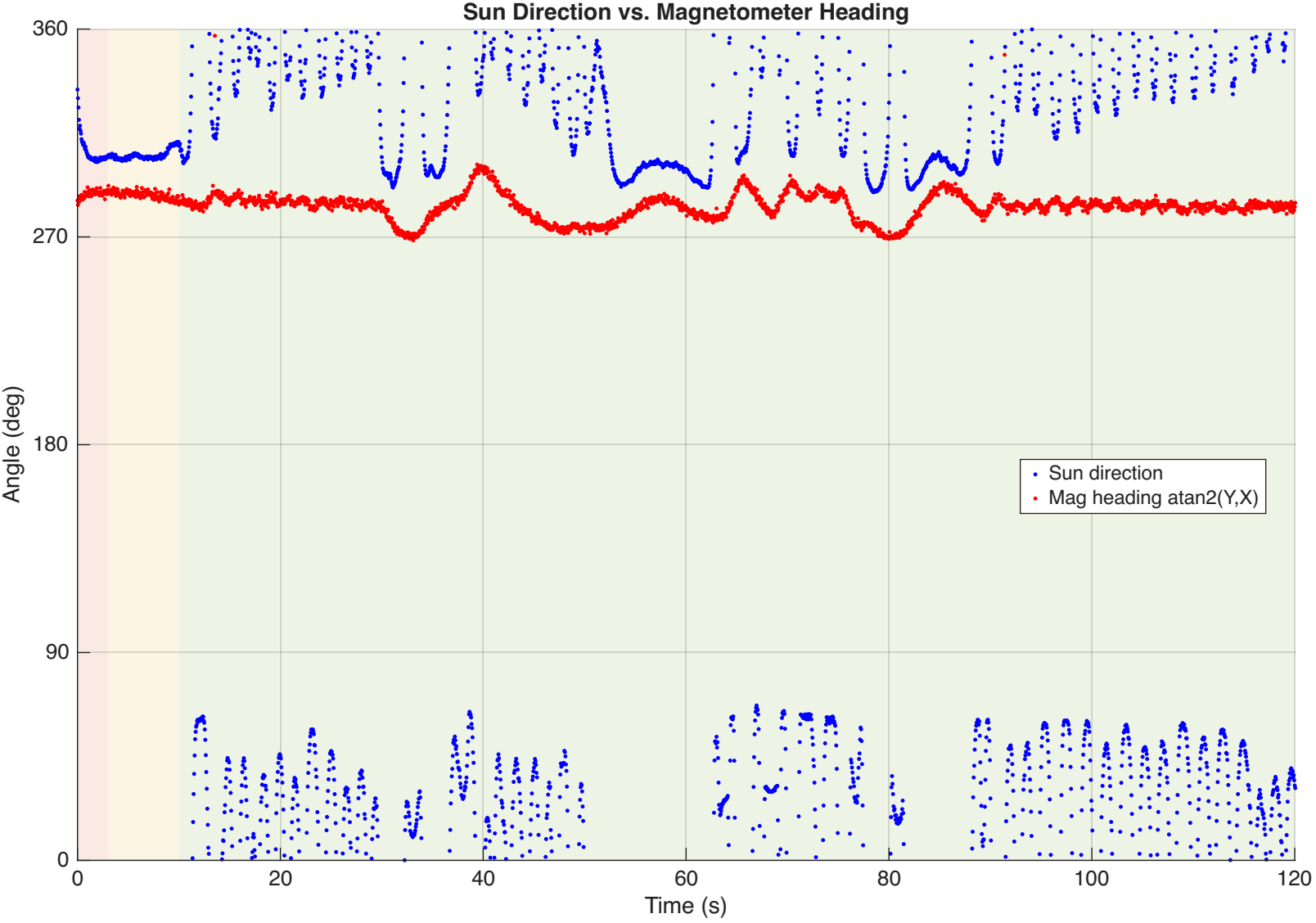

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
20
40
60
80
100
120
Pointing Error
Pointing error (deg)
180
135
90
45
0
-45
-90
-135
-180
SPINUP
HOLD
TRACK
Time (s)

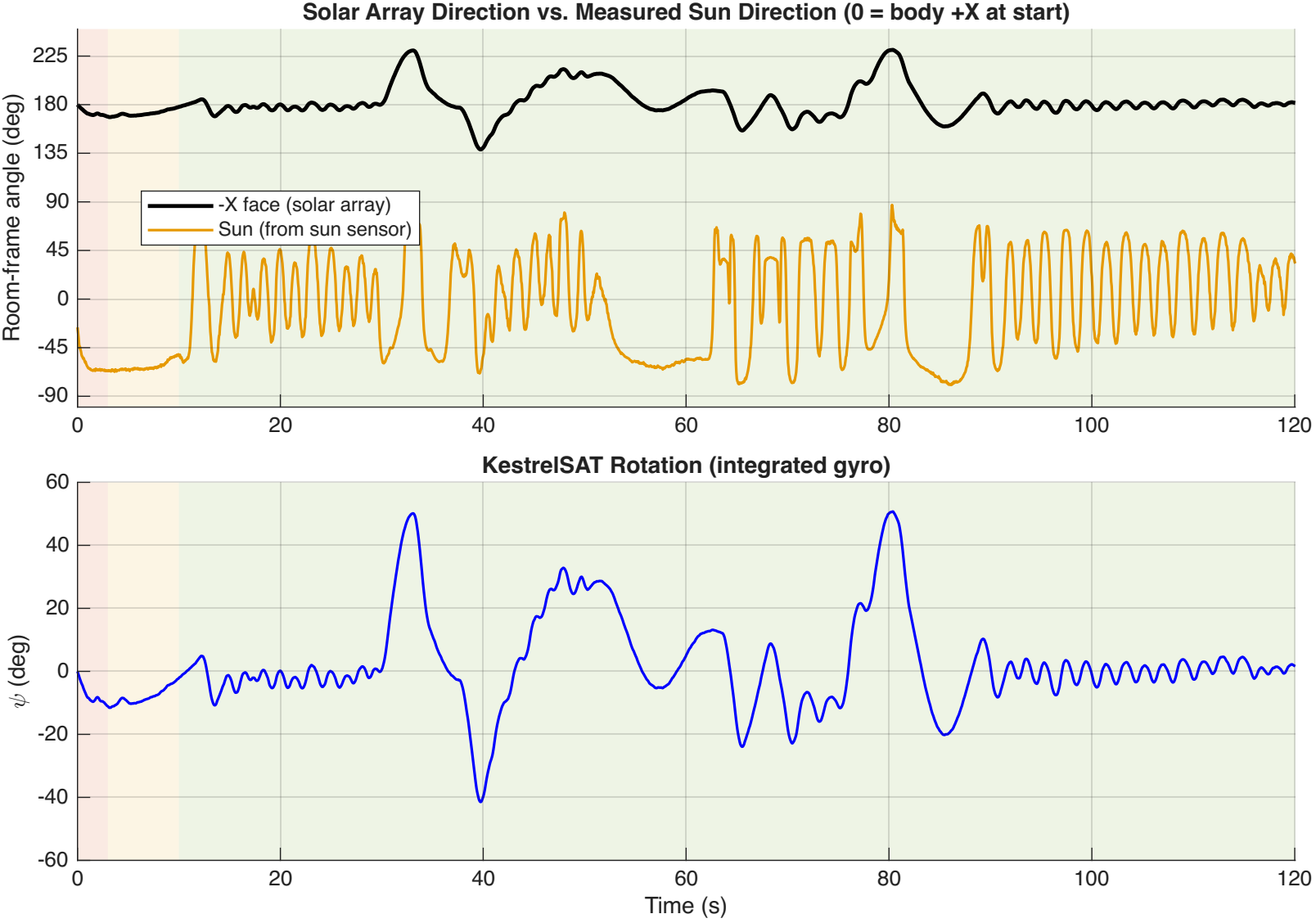

Z-Axis Body Rate (Gyro)
SPINUP
HOLD
TRACK
Body rate (deg/s)
Time (s)
40
30
20
10
0
-10
-20
-30
-40
0
20
40
60
80
100
120

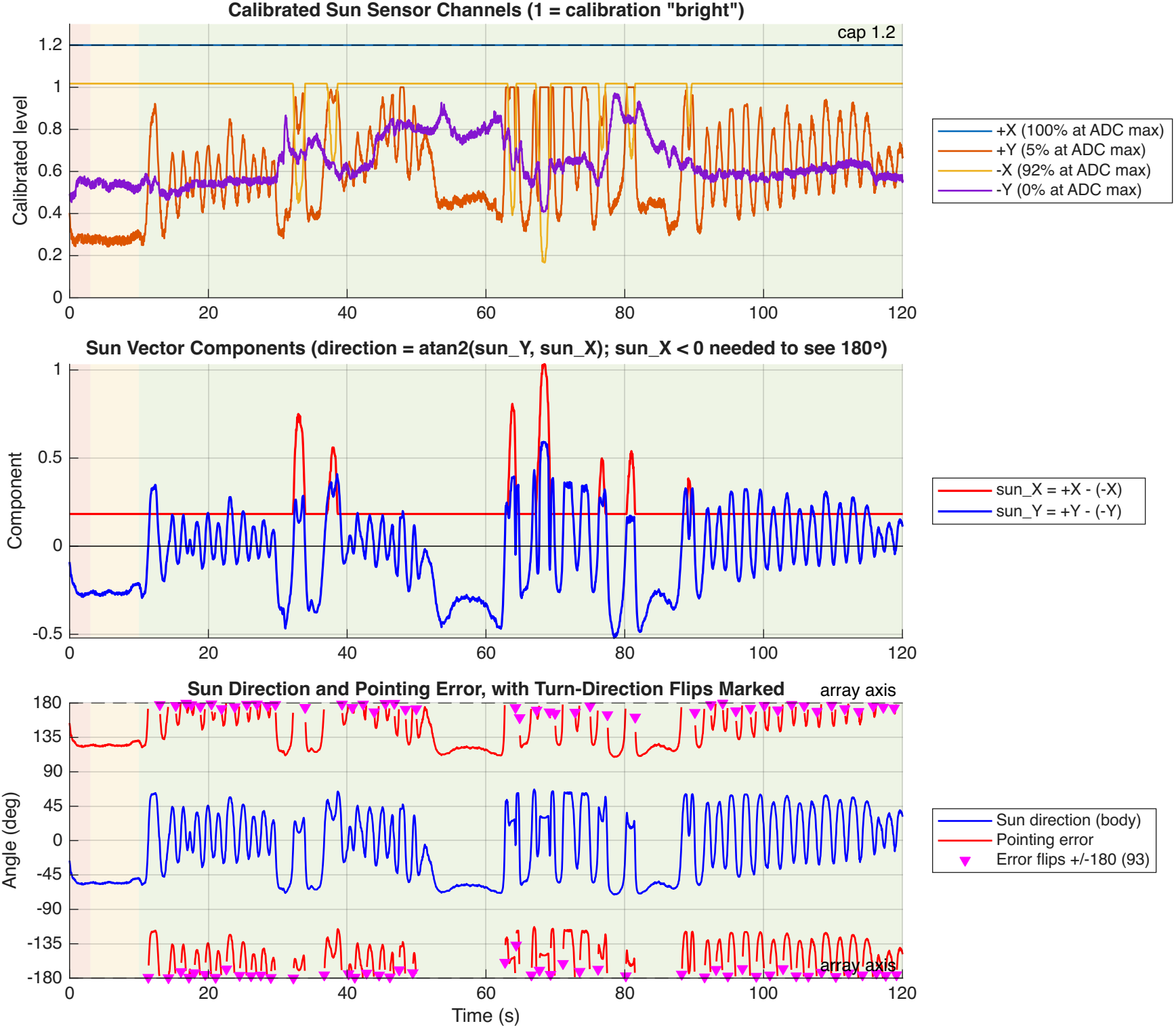

Reaction Wheel Speed
Wheel speed (RPM)
400
300
200
100
0
-100
-200
-300
-400
0
20
40
60
80
100
120
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
300
200
100
0
-100
-200
-300
Torque cmd
Saturated
Time (s)

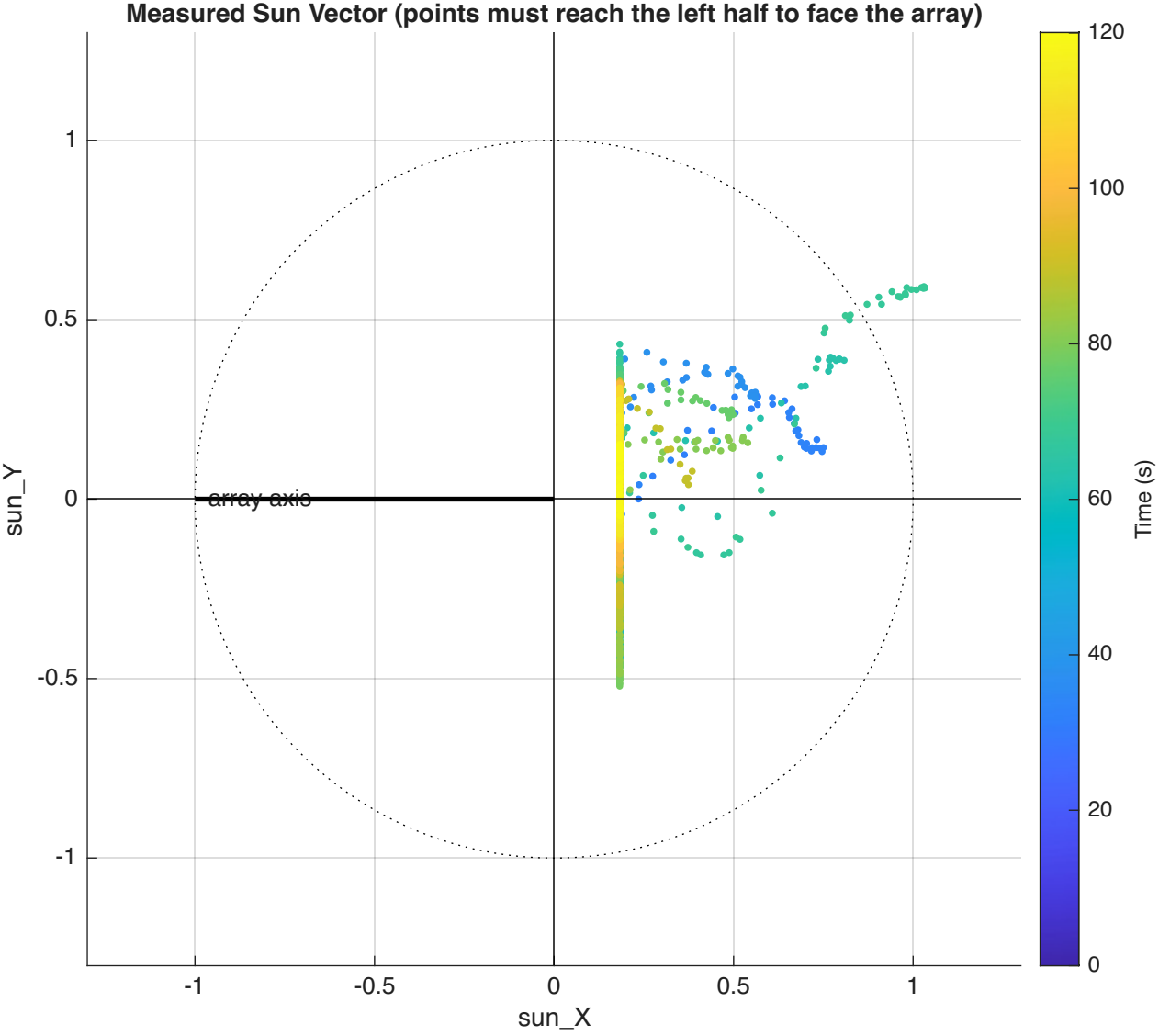

Sun Sensor Photodiode Readings
ADC max
ADC counts
4500
4000
3500
3000
2500
2000
1500
1000
0
20
40
60
80
100
120
+X
+Y
-X
-Y
Sun Vector Magnitude
sun_mag
4.2
4
3.8
3.6
3.4
3.2
3
2.8
2.6
Time (s)

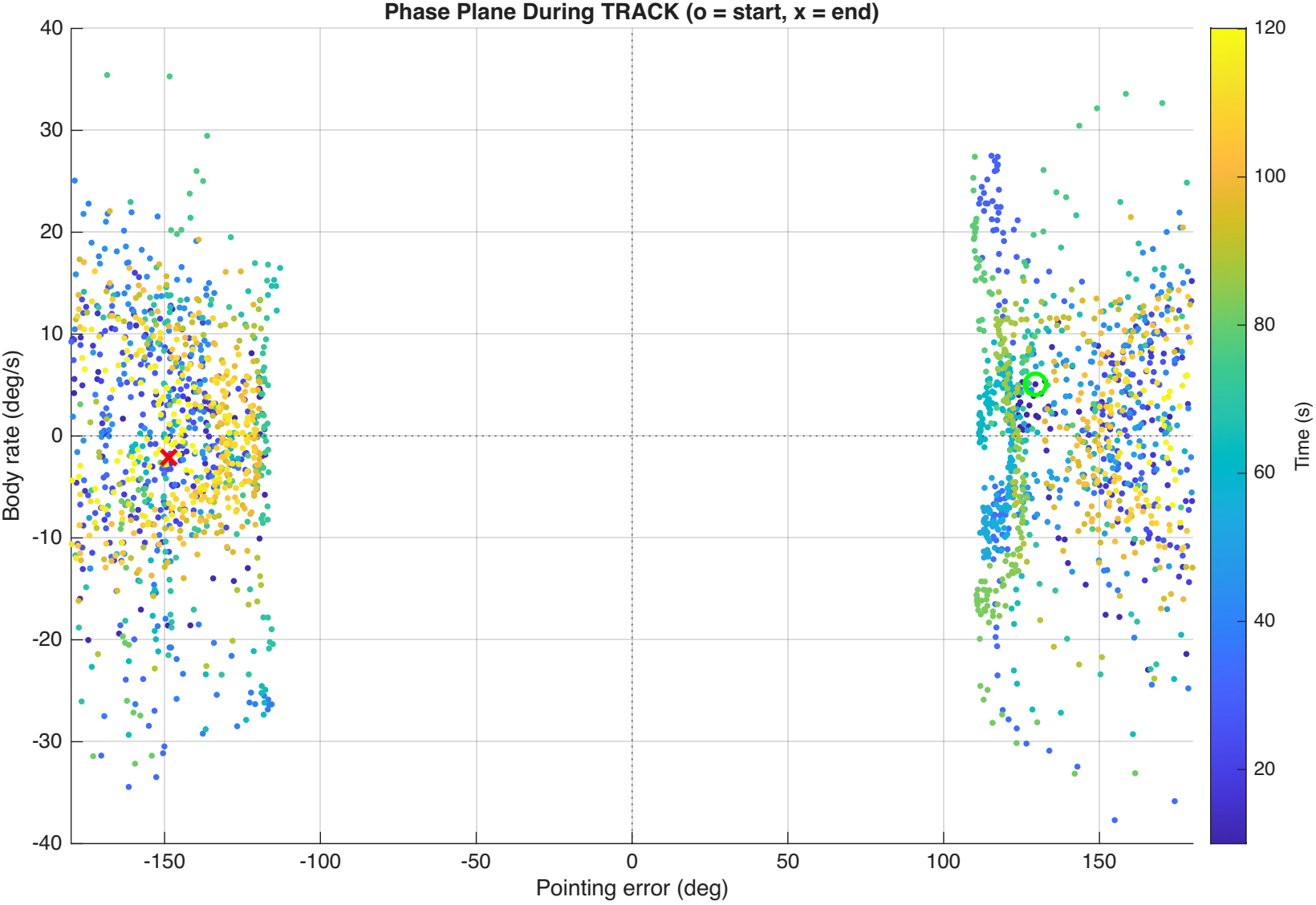

Magnetometer
Field (μT)
Time (s)
magX
magY
|Bxy|
Magnetometer X-Y (hard-iron check)
magY (μT)
magX (μT)
Time (s)

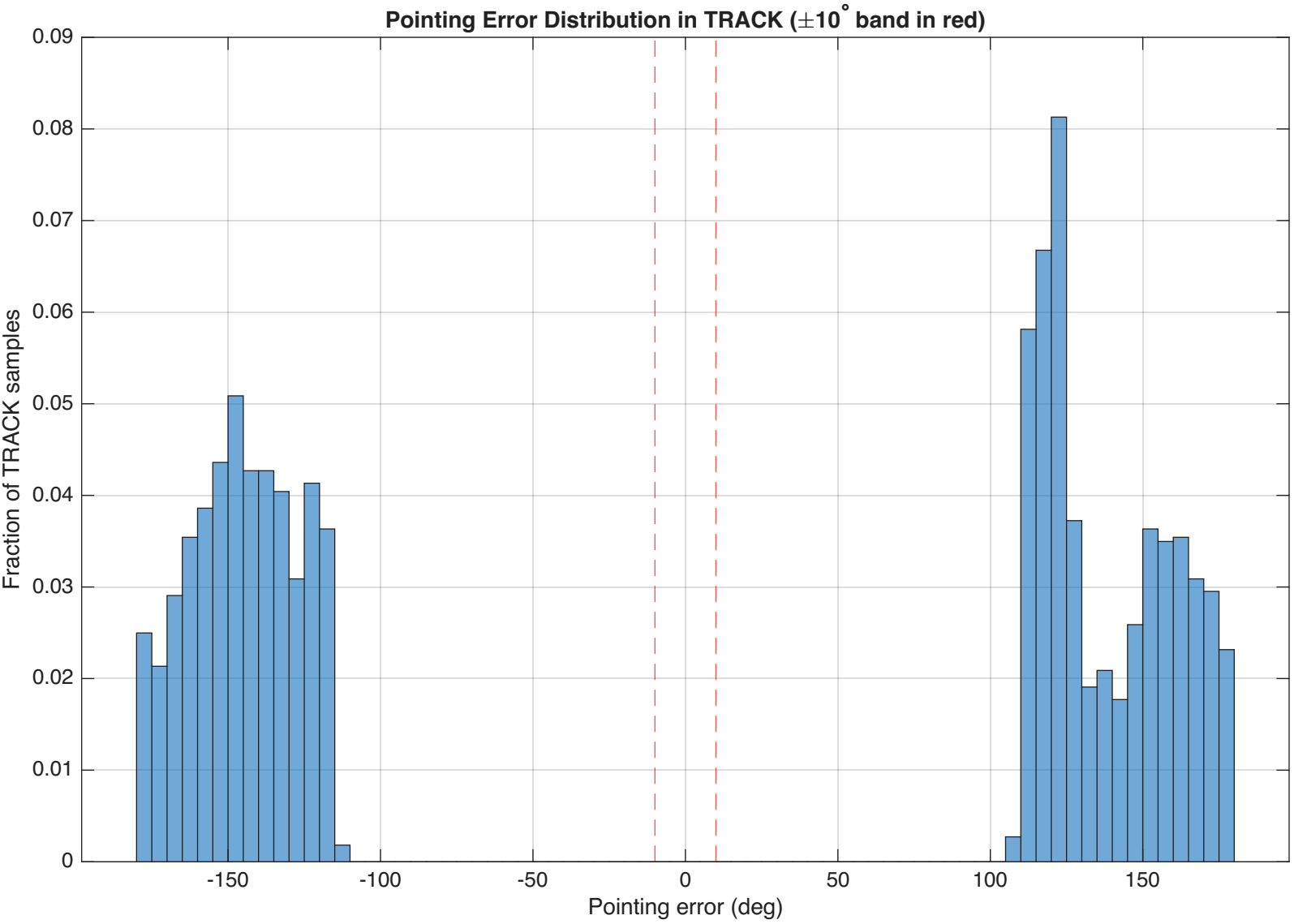

Sun Direction vs. Magnetometer Heading
360
270
180
90
0
Angle (deg)
0
20
40
60
80
100
120
Time (s)
Sun direction
Mag heading atan2(Y,X)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
Room-frame angle (deg)
225
180
135
90
45
0
-45
-90
-X face (solar array)
Sun (from sun sensor)
0
20
40
60
80
100
120
KestrelSAT Rotation (integrated gyro)
ψ (deg)
60
40
20
0
-20
-40
-60
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (100% at ADC max)
+Y (5% at ADC max)
-X (92% at ADC max)
-Y (0% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
array axis
Angle (deg)
Sun direction (body)
Pointing error
Error flips +/-180 (93)
Time (s)


Measured Sun Vector (points must reach the left half to face the array)
array axis
sun_X
sun_Y
Time (s)
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
120
100
80
60
40
20
0


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
40
30
20
10
0
-10
-20
-30
-40
-150
-100
-50
0
50
100
150
120
100
80
60
40
20


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
Pointing error (deg)
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
-150
-100
-50
0
50
100
150